MARCH 2025

RACKWARE MANAGEMENT MODULE

v7.4 - Prerequisites and Operational Requirements

RACKWARE INC

VERSION 3.49

Contents

1 First Time Customers - Workflow Overview 4
1.1 Creating ssh Keys 4
1.2 Installing the RMM 5
1.3 Creating the admin User 5
2 Workflow For Getting Started 6
3 Checklist of Prerequisites 6
4 Overview of RMM in the Network 8
4.1 RMM Directly Connected to Origin Server 8
4.2 RMM Connected Through Bridge Server 9
5 RMM Installation and Operational Requirements 10
5.1 RMM Server Requirements 10
5.1.1 Secure Boot not Enabled 12
5.2 OS Configuration and Network Configuration Requirements 12
5.2.1 Requirements during Installation of RMM 12
5.3 Backup/High Availability of RMM 13
5.4 RMM to Remain Powered On 13
5.5 Target Clouds/DataCenters Supported for AutoProvisioning 13
6 Supported Hypervisors, Operating Systems, Architectures, and FileSystems 14
6.1 Supported Target Hypervisor Environments 14
6.1.1 Supported Target Hypervisor Environments for Autoprovisioned Servers 14
6.1.2 Supported Target Hypervisor Environments for Preprovisioned Servers 14
6.2 Supported Origin Operating Systems 14
6.3 Supported Origin Architectures 15
6.4 Origin Servers in Good Working Order 15
6.5 Supported Origin FileSystems 15
7 Origin Host (Windows/Linux server) Setup Requirements 16
7.1 Firewall Ports Required to be Open 16
7.1.1 Firewall Ports Required to be Open on Servers 17
7.1.2 Firewall Ports Required if RMM GUI Being Used 17
7.1.3 Additional Firewall Ports Required to be Open when Autoprovisioning 17
7.2 Origin Host (Linux) Setup Requirements 18
7.2.1 Credentials 18
7.2.2 Free Extents 20
7.2.3 Utilities 21

7.2.4 Linux Antivirus Settings 21
7.2.5 No noexec on /tmp and /var/tmp 21
7.2.6 Miscellaneous 22
7.3 Origin Host (Windows) Setup Requirements 22
7.3.1 Volume Shadow Copy Service and Partition Free Space 22
7.3.2 Configuring Windows for SSH-Only 23
7.3.3 Windows Antivirus Settings 32
7.3.4 Windows Language Support 34
8 Miscellaneous Notes 34
8.1 Minimum Memory 34

1 First Time Customers - Workflow Overview

This document describes the workflow from purchasing the RMM to making it operational in your environment. It also describes the steps that should be taken with respect to your production servers and your network before installing the RMM to ensure that as soon as the RMM is installed you will be able to migrate or protect your production servers.

There are a few ways you may obtain the RMM:

- Via a Marketplace offering with a paid listing
- Via a Marketplace offering with a Bring Your Own License (BYOL) listing
- Independent of a Marketplace offering

The initial steps to install and license the RMM vary depending on how you obtained it.

If you obtained the RMM through a Marketplace offering with a paid listing, then the RMM is already installed and already licensed. However, the RMM likely does not have the ssh keys needed and does not have the admin user created. Follow the steps in Creating ssh Keys to create the necessary ssh keys.

If you obtained the RMM through a Marketplace offering with a BYOL license, then the RMM is already installed but not yet licensed, the admin user needs to be created, and likely does not have the ssh keys needed. Follow the steps in Chapter 2 of the RMM v7.4 Installation Guide to obtain a license and the steps in Creating ssh Keys to create the necessary ssh keys.

If you are getting the RMM outside of a Marketplace offering, then the RMM has not yet been installed or licensed. Once it is installed it will have the ssh keys needed and the admin user will be created. Follow the steps in Chapter 1 of the RMM v7.4 Installation Guide to install the RMM and in Chapter 2 of the RMM v7.4 Installation Guide to obtain a license.

1.1 Creating ssh Keys

If you have deployed your RMM from a Marketplace listing, you may find that your RMM does not have a public key, in which case you will not be able to set up your Linux and Windows source servers such that the RMM can ssh to them.

After deploying an RMM from a Marketplace, from the RMM's console window, issue the command "sudo su-" so you have root access, then issue "cd /root/.ssh", followed by "ls".

```
[root@RMM01 ~]# cd /root/.ssh
[root@RMM01 .ssh]# ls
      authorized_keys  known_hosts
```

If the files id_rsa.pub and id_rsa are not shown, then generate them by issuing the command "ssh-keygen". Press the <enter> key each time there is a prompt.

```
[root@RMM01 .ssh]# ssh-keygen
      Generating public/private rsa key pair.
      Enter file in which to save the key (/root/.ssh/id_rsa):
      Enter passphrase (empty for no passphrase):
```

```
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa.
Your public key has been saved in /root/.ssh/id_rsa.pub.
The key fingerprint is:
SHA256:qvjkOCSOXCs3VeipsgND1dvtH5liZ0R0j22jKQzZtjU root@us-
east-default-vpc-bob3vsi
The key's randomart image is:
+---[RSA 2048]----+
|    .     .. .   |
|   . .    o.. +  |
|  .   + .o.o E = |
| .   o o .+.o = .|
|.   . o S .+oo   |
|+ .. + . + *.    |
|=+. = . . = .    |
|.*.X .     .     |
| .Oo=            |
+----[SHA256]-----+
```

[root@RMM01 .ssh]#

Now, while still in the .ssh directory, issue the "ls" command again.

[root@RMM01 .ssh]# ls

```
authorized_keys  id_rsa  id_rsa.pub  known_hosts
```

Now the id_rsa and id_rsa.pub files are shown.
The id_rsa.pub file contains the RMM's public key. The contents of this file is used in the when setting up the prerequisites for Linux hosts and Windows hosts as described in Credentials (LInux) and Configuring Windows for SSH-Only.

Do not rename the id_rsa and id_rsa.pub files. Doing so will cause ssh'ing to the source servers to not work properly.

1.2 Installing the RMM

If the RMM was not obtained from a Marketplace listing, it will need to be installed. Contact RackWare Support at support@rackwareinc.com and let them know if you have purchased the RMM or if you are interested in a Proof of Concept RMM license. The Support team will reply with instructions for downloading the appropriate RMM installer

1.3 Creating the admin User

The 'admin' user is used to access the RMM GUI. If the RMM was obtained from a Marketplace listing, you will need to add this user and generate a password for it.

If you do not already have root access to the rmm, issue the command "sudo su-" to get root access. Then issue the commands:

```
useradd admin
passwd admin
```

The password must meet the password requirements of the operating system on which the RMM is running.

2 Workflow For Getting Started

Unless you have purchased a Professional Services contract (in which case someone from RackWare will be performing all of your migrations and DR protections), or you have installed the RMM from a Marketplace, once you have purchased licenses for the RMM, or reached an agreement for a Proof of Concept installation, please contact licensing@rackwareinc.com and ask for credentials for the RackWare ftp server (ftp://ftp.rackwareinc.com) so that you can download the appropriate RMM installation file.

If you have purchased licenses, you can also contact RackWare Support via an email to support@rackwareinc.com to have an account set up for you on the RackWare support site at https://rackware.freshdesk.com. This site contains the latest RMM release notes, documentation, and a knowledgebase of solutions for common issues.

If you have gotten the RMM from one of the cloud Marketplaces, follow the instructions from the cloud documentation for installing the RMM. If you did not get the RMM from one of the cloud marketplaces, then the RMM installation file will be placed on your section of the RackWare ftp server. Please then download the RMM code from the RackWare ftp server and install it on your RMM server following the instructions in the Rackware RMM v7.4 Installation Guide .

At the end of the installation a file of the form rwlicense_preinstall_nnnnnnnn in the /etc/rackware/ directory will be created. Please email that file to licensing@rackwareinc.com. RackWare will then generate a license and send it to you along with instructions for installing the license. You can then install the license and begin migrating/protecting your servers. Chapter 2 of the Rackware RMM v7.4 Installation Guide contains more information about the RackWare licenses.

3 Checklist of Prerequisites

The Prerequisites include specifications for the RMM server itself and for the RMM's environment, and specifications for the servers that will be migrated/protected.

Please note that "origin servers" or "source servers" are the servers in the production environment while the "target servers" are the servers in the cloud environment. Migration is occurring *from* the origin/source servers *to* the target servers.

The list below shows a summary of the prerequisites for the RMM VM itself and for the origin servers. For each prerequisite the secont of this document where the details of the prerequisite can be found is shown.

The following Prerequisites apply to the RMM Server itself:

- Target network isolated from Source network (sections 4.1 and 4.2)
- RMM is supported on the OS of the RMM Server (section 5.1)
- CPU and RAM planned on the RMM server will be adequate (section 5.1)
- Space planned for the RMM server will be adequate (section 5.1)
- zfs storage required will be available (section 5.1)
- RMM Server meets the OS Configuration and Network Configuration requirements (section 5.2)
- Port 443 will be open (outbound from the RMM) during RMM installation (section 5.2.1)
- RMM will have access to a working yum manager that has access to an EPEL package during RMM installation (section 5.2.1)
- RMM will have a method of being backed up (section 5.3)
- RMM to remain powered on if BYOL licenses being used (section 5.4)
- If using auto-provisioning, target cloud is on list of supported clouds (section 5.5)
- Recent version of Chrome, Firefox, or Edge browser available ot use for the Graphical User Interface (GUI) of the RMM
- RMM must not have Secure Boot enabled (section 5.1.1)

The following prerequisites apply to all origin servers:

- The server must be running an OS that is on the list of supported Oses (section 6.2) and have an architecture that is on the list of supported architectures (section 6.3)
- The file systems on the server must be on the list of supported file systems (section 6.5)
- Any WAN links from the origin environment to the RMM must have a capacity of at list 100 megabits/sec available for use by the RMM
- Any links from the RMM to the target servers must have a capacity of at least one gigabit/second available for use by the RMM
- Port 443 must be open from the RMM to the target cloud (section 7.1)
- An ssh port (usually port 22) must be open from the RMM to the orgin servers (section 7.1)
- An ssh port (usually port 22) must be open from the RMM to the target servers (that is, to the servers in the destination cloud) (section 7.1)
- If using autoprovisioning/dynamic provisioning, an ssh port (usually port 22) must be open from the RMM to the target cloud (section 7.1)
- Required ports must be open on any intermediate firewalls (section 7.1)
- All logical volumes must have unique file system identifiers (UUIDs or labels) (section 6.5)

The following prerequisites apply to Linux origin servers:

- The RMM's public key must be copied to the authorized_keys file of the origin server (section 7.2.1)
- If a user other than 'root' will be used by the RMM to access the origin server, then update the /etc/sudoers file on the origin server (section 7.2.1)
- Verify that each volume group on the origin server has at least 15% of the Used Space of the volume group available as free extents (section 7.2.2)
- If target servers will be using PXE-boot, verify the required utilities are on the origin server (section 7.2.3)
- Whitelist /mnt/rackware on any antivirus product on the origin server (section 7.2.4)
- There must be an /etc/default/grub file on the origin server (section 7.2.6)
- There must be at least 20 MB free space in /var/tmp (section 7.2.6)
- /tmp and /var/tmp must not have noexec specified in /etc/fstab (section 7.2.5)

The following prerequisites apply to Windows origin servers:

- Port 445 must be open from the RMM to the target cloud (section 7.1)
- Shadow copy storage space should be at least 15% of the Used Space for each volume (section 7.3.1)
- Free space on the OS drive (typically the C drive) must be at least 1 GB (section 7.3.1)
- The server has the RackWare MSI package installed and running on it (section 7.3.2)
- Whitelist, in any antivirus product on the server, the required files and directories (section 7.3.3)
- Check Windows Language Support (section 7.3.4)
- Port 443 must be open from the RMM to the target cloud and inbound to the RMM (section 7.1)

4 Overview of RMM in the Network

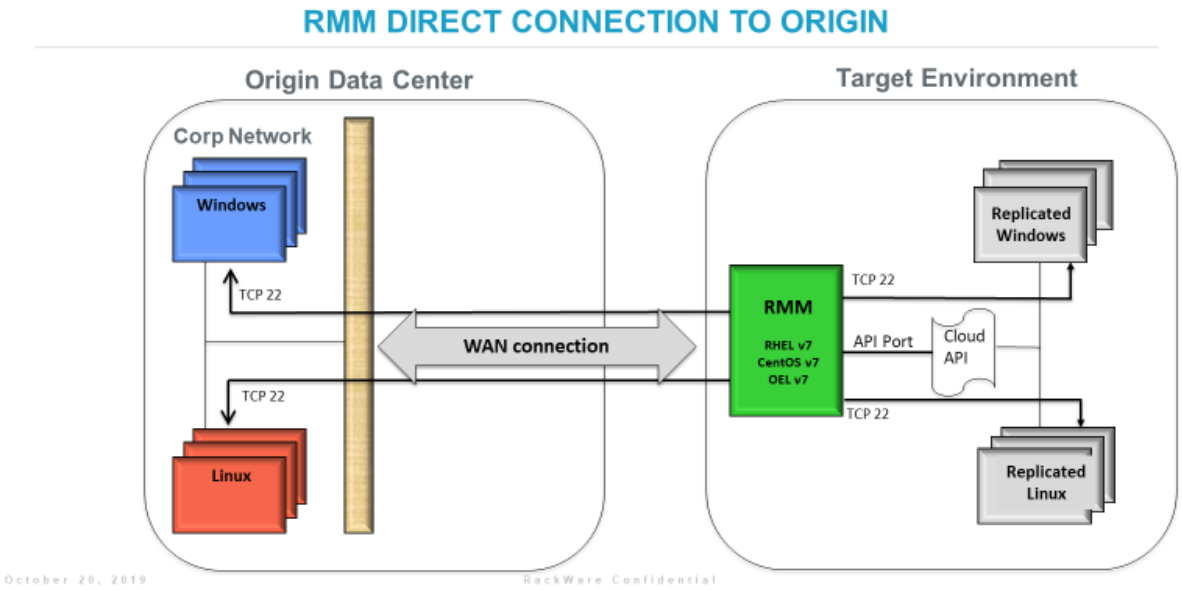
The Rackware Management Module (RMM) is a Linux application and runs on a RedHat, CentOS, or Oracle Linux installation. The RMM does not require any interface to a hypervisor or storage array. The RMM connects to the Origin server over the network at the Operating System level and replicates the server Image. The Image can be replicated to a storage location at the Target site or to a provisioned server (virtual or physical) at the Target site. The RMM is also capable of auto-provisioning appropriately sized servers in the Target environment.

A Configuration Management Data Base (CMDB) is maintained by the RMM on the RMM Server. The CMDB keeps track of resources the RMM is managing, optionally captured Images, as well as operational state

At a high level, the RMM can be in two types of network topologies. The RMM server can be directly connected (via a TCP connection) to the Origin server or the RMM server can be connected to the Origin server via a Bridge Server which provides an IP NAT function.

4.1 RMM Directly Connected to Origin Server

The most common configuration is to have the RMM establish a TCP connection directly to the Origin server. The diagram below shows the topology and ports that need to be opened when the RMM is performing a Capture/Sync operation to a physical or virtual server in the Target environment. Please see section 7.1 for further information about the ports that are required to be opened.



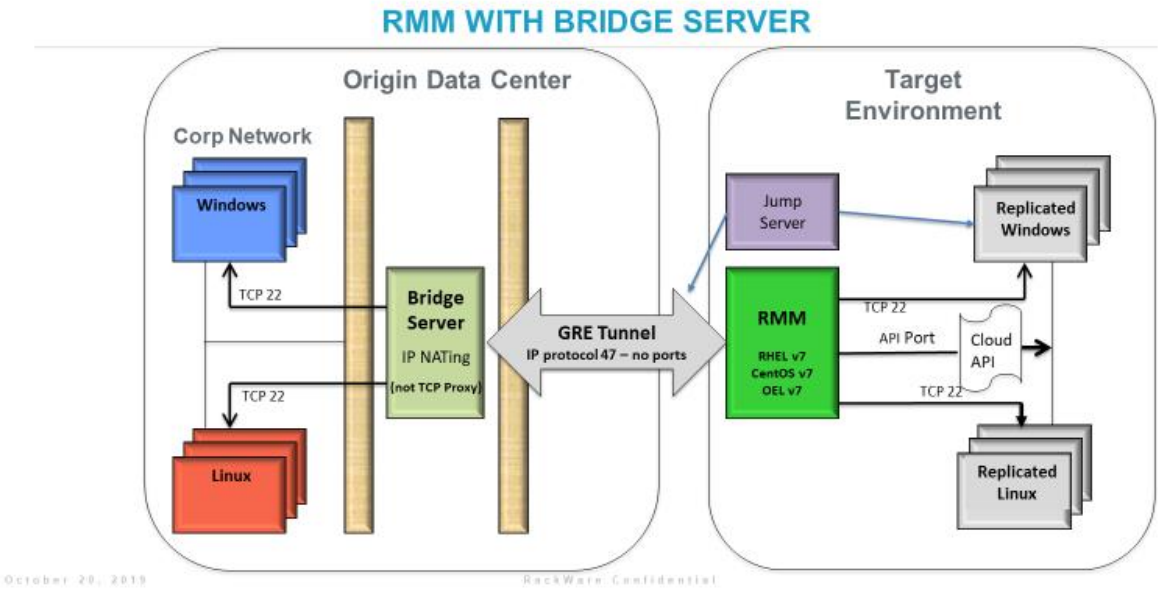
Note that the network containing the origin/source servers is isolated from the network containing the target servers – the only communication path between the 2 networks is through the RMM. The origin/source network must always be isolated from the target network. If they are not isolated, when the target servers are booted into their OS, Active Directory may confuse a source server with a target server which could cause issues in the source network.

4.2 RMM Connected Through Bridge Server

If you wish to have the addresses of the Origin servers be the same as the addresses of the Target servers, one way to do this is by using an RMM Bridge Server. The RMM Bridge Server acts like a proxy; the iptables rules set up will forward packets directly to the RMM tunnel IP. It is also capable of performing the NAT translations.

Please note that there is no specific “bridge server” package provided by RackWare. The Bridge Server is a standard Linux server with standard Linux networking files configured in a specific way so that it acts as an RMM Bridge Server.

The diagram below shows the topology and ports that need to be opened when the RMM is performing a store and forward replication to a physical or virtual server in the Target environment via an RMM Bridge Server.



Please note that a Bridge Server could also be deployed in a cloud environment, in which case its configuration and port requirements could be different than what is shown above.

Note that the network containing the origin/source servers is isolated from the network containing the target servers – the only communication path between the 2 networks is through the RMM. The origin/source network must always be isolated from the target network. If they are not isolated, when the target servers are booted into their OS, Active Directory may confuse a source server with a target server which could cause issues in the source network.

5 RMM Installation and Operational Requirements

5.1 RMM Server Requirements

Installation set up for RackWare is comprised of a dedicated server (physical or virtual) running the RMM. The RMM server requirements are as follows:

- x86_64 architecture (Intel or AMD)
- RHEL / CentOS / OL v7 (7.4 to 7.9) or RHEL/OL/RockyLinux/AlmaLinux 8.x
- 70 GB storage for /
- 12 GB or more of storage on /tmp
- 6 GB or more of storage on /srv
- 25 GB or more of storage at /opt
- 15 GB or more of storage at /var/log
- 32 GB memory
- 4 cores (vCPU)
- An additional disk for zfs storage***

Please note that having separate partitions for /opt, /srv, /var/log and /tmp is not required - there can be a single "/" partition that contain /opt, /var/log, /tmp, and /srv as long as there is enough free space to meet all of the space requirements. In the case of a single / partition, a minimum of a 70 GB / partition should be used.
If separate mount points are set up for /opt, /srv, /var/log, or /tmp, be sure to set up those mount points in the /etc/fstab file.
To determine the number of servers an RMM in a given cloud can support, see the RMM Sizing Considerations section for your particular cloud in the 7.4 RMM GUI Operations Guide. The average RPO time of the servers must be known. Cpu/core definitions differ between clouds. A 'vcpu' in some clouds contain half a core.

As a guideline, a generic server configured as described above can support the following in a DR (Staged Syncs) configuration:

Average RPO time of the servers being protected	Number of Servers being protected on an RMM with a CPU containing 4 cores/32 GB RAM
< = 1 hour	Up to 20
2 hours	Up to 25
3 hours	Up to 30
4 hours	Up to 33

Note that the amount of memory needed will vary depending on the actual workload, such as if the sync contains a very large number of files or very large files. Some workloads will require more memory, others will require less memory.
When deploying RMMs for a pure migration solution with Staged Syncs, see the subsection of Chapter 5 of the RMM 7.4 GUI Operations Guide appropriate for the cloud in which your RMM resides. A generic RMM built with 4 cores and 32 GB RAM will allow you to do up to 20 simultaneous captures or syncs if using Staged Syncs.

When deploying RMMs for a pure migration solution with Host Syncs (with passthrough), see the subsection of Chapter 5 of the RMM 7.4 GUI Operations Guide appropriate for the cloud in which your RMM resides. If Host Syncs (with passthrough) are being used for a pure migration solution, as a guideline, a generic RMM built with 4 cores and 32 GB RAM will allow you to do up to 24 simultaneous syncs.

*** Regarding the zfs storage requirements:

- if you will be doing staged syncs (also known as store and forward syncs), then a raw, unformatted disk device(s) equal in size to the total used space of the production servers should be available at the time of the RMM installation. During the installation procedure the RMM will ask for the name of this disk(s), so it should be available before installing the RMM software.
- The storage must be block storage - not NFS or CIFS storage
- If you are planning on using the RMM for Disaster Recovery then our best practice is to use Staged Syncs, which means zfs storage will be needed.
- If you are planning on using the RMM for Backup then the ZFS space needed will be the space needed for the staged syncs plus the space needed for the backed up images.

5.1.1 Secure Boot not Enabled

The VM on which the RMM is to be installed must not have Secure Boot enabled. Secure Boot must be either disabled or not supported on the VM. The command "/usr/bin/mokutil --sb-state" will show the status of Secure Boot.

5.2 OS Configuration and Network Configuration Requirements

The RMM installation requires the following additional configuration:

- Working yum manager with EPEL package
- Access to an NTP server
- Access to an SMTP server (can be a proxy)
- SELinux is disabled (please contact RackWare if you need this to be enabled)
- Name resolution must be configured and working.
- Install as root; if root access is not available on the RMM, a userid in the sudoers file may be used as long as the userid has" ALL=(ALL) ALL"permissions in the sudoers file.
- The root user must be defined in the sudoers file.

5.2.1 Requirements during Installation of RMM

The requirements in this section apply only during the installation of the RMM. Once the RMM has been successfully installed these requirements are no longer applicable.

The following port must be opened on any firewalls from the RMM server to the Internet:

TCP port 443

The RMM must have access to a working yum manager with access to an EPEL package

As described in the RMM 7.4 Installation Guide, before doing a first time installation of the RMM software, a 'yum update' will be executed.

If using RHEL without a subscription, a DVD should be configured as the repo.

In the majority of instances for a standard RedHat/CentOS install the following domains will be queried. As a security measure it is possible to limit Internet connectivity by whitelisting these domains:

download.fedoraproject.org
dl.fedoraproject.org
mirror.centos.org
archive.zfsonlinux.org

If the RMM is being installed on a VMware installation also add "packages.vmware.com" to the whitelist

If for some reason these domains are not sufficient, on the RMM Server, execute:

```
cat /etc/yum.repos.d/epel.repo | grep -i http
```

This will iterate the domains that are being used for your particular configuration.

For licensing information see the Installation Guide. Note that once licenses are applied if the RMM is powered down for more than 8 days the licenses will be void. If it's necessary to power down the RMM for more than 8 days, please contact RackWare Support.

5.3 Backup/High Availability of RMM

RackWare software runs on a Virtual Machine in a Target Cloud environment. There are two components of storage that need to be protected in the unlikely, but possible, case that a fatal failure occurs in either the VM itself or the associated storage. There are the images themselves and a CMDB that retains configuration information, server specifications, and operational state information.

On the RMM, all storage is attached block storage. From a resiliency and recovery perspective, a newly provisioned RMM is designed to quickly and easily import any copy, clone, or backup of the CMDB and Image storage to resume operations. In a recovery scenario, upon importing the CMDB, the RMM returns all system operations and policies to the correct desired execution and state. The RMM that imports the CMDB and Images can be newly provisioned or pre-provisioned as a hot standby. Given the speed at which Clouds can spin up new servers it's usually most economical to simply provision the new VM and install the RMM software upon a recovery operation.

Almost all Cloud Service Providers have disk cloning or snapshot capability for all provisioned block storage. While more elaborate and expensive resiliency systems can be configured, RackWare recommends utilizing the Cloud disk snapshot services. In the unlikely event of a fatal failure, the disk snapshot can be used to recover both the data and the operational state.

5.4 RMM to Remain Powered On

The RMM should remain powered on after it is deployed. If the RMM is powered off for more than 8 days, its license will expire, and a new license will be needed. Contact licensing@rackwareinc.com if this occurs.

5.5 Target Clouds/DataCenters Supported for AutoProvisioning

While the RMM can migrate servers to any cloud, if you wish to use the RMM to autoprovision servers into a cloud, then the cloud must be on the list of clouds that the RMM supports for autoprovisioning. This list includes:

Oracle Cloud (OCI)
Azure
Google Cloud Platform (GCP)
vCenter/OCVS
Zadara Cloud
IBM Cloud VPC
Oracle PCA X9-2, C3
OVHcloud
Amazon Web Services (AWS)
Digital Ocean
Linode

6 Supported Hypervisors, Operating Systems, Architectures, and FileSystems

6.1 Supported Target Hypervisor Environments

The set of supported hypervisor environments in the target environment varies depending on if the target servers are being autoprovisioned or if they are being pre-provisioned.

6.1.1 Supported Target Hypervisor Environments for Autoprovisioned Servers

The RMM supports autoprovisioning servers into the following Target hypervisor environments:

VMware ESXi
KVM
Oracle VM
Oracle Linux Virtualization Manager

This assumes that the OS of the source server is supported, and that the target cloud is supported for autoprovisioning.

6.1.2 Supported Target Hypervisor Environments for Preprovisioned Servers

The following target hypervisor environments are supported when the target servers are preprovisioned:

Hyper-V
Nutanix
RHEV
Xen
Any hypervisor that is supported with autoprovisioned servers

Please contact info@rackwareinc.com if support for any other hypervisor is desired.

6.2 Supported Origin Operating Systems

RackWare supports migrating and protecting servers running the following Linux distributions and Windows OSes:

RHEL 5.2 through 5.11,6.x, 7.x, 8.x, 9.x
CentOS 5.2 through 5.11, 6.x, 7.x, 8.x
Oracle Linux 5.6 through 5.11, 6.x, 7.x, 8.x, 9.x

SLES 11, SLES 12 (no btrfs)*

SLES 15 (no btrfs)*

Ubuntu 10 12 14 16 18, 20 22 24	Debian 8,9,10,11
AlmaLinux 8,9	Rocky Linux 8,9
Windows Server 2008 R2**	Windows Server 2012
Windows Server 2012 R2	Windows Server 2016
Windows Server 2019	Windows Server 2022
Windows 7	Windows 10

For any OSes in the above list which have a 32-bit version, the 32-bit version is supported as well as the 64-bit version.
The servers may be using UEFI or BIOS with any of the above OSes, except that RHEL/Centos/OEL 5x servers are not supported if using UEFI.

RackWare provides support for migrating Active Directory/Domain Controller servers under certain circumstances. Contact RackWare for more information. Please note that in general, RackWare is not responsible for configuring or setting up Active Directory or Domain Controllers in the target environment.

If support for any other OS is desired, contact info@rackwareinc.com

Please note that the RMM supports migrating servers to a cloud only if the server is running an OS for which the cloud has an image.

* syncing btrfs file systems is not supported.

** Most Cloud Service Providers have ended support for Windows 2008. We recommend that you upgrade your source servers to a later version of Windows before migrating them into the cloud. In most cases migrations of Windows 2008 R2 will continue to work and RackWare will continue to support these Migrations with the understanding that there are no guarantees that this version of Windows will work in the target Cloud, and support for such systems in target Clouds is at the discretion of the Cloud Service Provider.

Windows 2008 R1 is no longer supported by RackWare.

6.3 Supported Origin Architectures

The source servers must have an x86 or x86_64 (Intel or AMD) architecture.

6.4 Origin Servers in Good Working Order

Any servers being migrated must be in good working order. It cannot have any major problems that prevent the OS from working as designed (snapshots need to work, package manager needs to be in working order, can't be missing critical components, etc). The server needs to reboot without any issues, and if Windows Updates are configured on the server, the server must be able to successfully install the updates.

6.5 Supported Origin FileSystems

The RMM supports the following Linux filesystems when Capturing from, and Assigning to a local disk:

ext2	ext3
ext4	GFS2
ReiserFS	XFS
FAT	

Origin/Source Hosts must be running `LVM` or `device mapper`. The RMM also supports `vxfs`; contact RackWare for more information.

The RMM supports NTFS and FAT filesystems for Windows when Capturing from, and Assigning to, a local disk. ReFS filesystems for Windows are also supported – ReFS filesystems on a source server will be converted by the RMM to NTFS filesystems on the target servers.

The RMM does not currently support migrating Windows volumes that have deduplication turned on or that use Microsoft Storage Spaces.

NFS and CIFS file systems are also supported when they are implemented on Linux or Windows servers. In this context they are just another server that is fully supported.

For NFS/CIFS appliances where the RMM does not have access to the appliance APIs the RMMs only access is from the Client side. Access from the client side does not have any snapshot capability so replication of NFS/CIFS from a Client cannot guarantee consistency unless the applications are quiesced. Hence migration of these NFS/CIFS filesystems can be accomplished when the application is quiesced. These NFS and CIFS environments are further complicated when implementing multi-writer configurations. DR from the Client side is not supported as it's impractical to quiesce the application for each sync. It may be possible to support DR in these environments when provided access to the appliance API, but this needs to be evaluated for compatibility with the Target NFS/CIFS implementation.

All file systems should have unique file system identifiers (UUIDs, serial numbers, or labels)

7 Origin Host (Windows/Linux server) Setup Requirements

This section describes the firewall ports that must be open on the servers and in the clouds, the firewall ports that must be open in the cloud when performing autoprovisioning, and the required configuration items on the Linux and Windows servers that will be migrated or protected by the RMM.

7.1 Firewall Ports Required to be Open

The RMM utilizes a few tcp ports during migration disaster recovery operations. The ports that are required to be opened are described in the table below and in the next 3 subsections.

Port Number	Direction	When	Additional Notes

22	From RMM to Source server	All migrations/DR/Backup	See Section 7.1.1
22	From RMM to Target server	If using pre-provisioned targets	See Section 7.1.1
22	From Source server to Target server and From Target server to Source server	If using Direct Syncs without the passthrough option	See Section 7.1.1
443	From whatever hosts are running web browser to RMM	If using RMM GUI	See Section 7.1.2
22	From RMM to Target cloud	If autoprovisioning servers	See Section 7.1.3
445	From RMM to Target cloud	If autoprovisioning Windows servers	See Section 7.1.3
443	From RMM to Target cloud	If autoprovisioning servers	See Section 7.1.3

7.1.1 Firewall Ports Required to be Open on Servers

A port for SSH must be open for incoming traffic on the origin/source server and on any firewalls between the RMM and the source server. Typically, this is port 22 - by default, the RMM uses port 22 as the ssh port on an origin server. So port 22 is the port that must be open for inbound traffic on the origin server and on any firewalls between the RMM and the origin server.

If, however, port 22 is being used by some other application on an origin server, then the RMM can use a different port on the origin server as the ssh port, known as a "custom ssh port". Whichever port is being used as the ssh port on the origin server is the port that must be open on the firewalls. The target server will use the same custom ssh port as the origin server.

The ssh port is the only port that needs to be open in the inbound direction on the origin servers.

If you will be pre-provisioning a target server, then the ssh port must be open in the inbound direction on the target server.

If you will be using direct syncs - syncing from a source server to a target server without the traffic passing through the RMM - then the ssh port must be open in both directions on the source server and in both directions on the target server. Typically syncs between hosts are done with the traffic passing through the RMM.

7.1.2 Firewall Ports Required if RMM GUI Being Used

If the RMM GUI will be used, then Port 443 must be open inbound to the RMM from whatever host/hosts will be running the RMM GUI.

7.1.3 Additional Firewall Ports Required to be Open when Autoprovisioning

When autoprovisioning a server to a cloud, port 22 must be open from the RMM to the cloud.

In addition, when autoprovisioning a Windows server to a cloud, port 445 must be open from the RMM to the cloud unless a custom image is being used. The only exception is if the "cloud" is VmWare/vCenter. In that case port 445 does not need to be open from the RMM to VmWare/vCenter when autoprovisioning to VmWare/vCenter.

Any ports that a cloud API or a hypervisor requires must be open from the RMM to the cloud/hypervisor. For many clouds (e.g. OCI, GCP) the port that must be open from the RMM to the cloud/hypervisor is port 443. For Oracle PCA, port 443 must be open from the RMM to the PCA Management Node.

7.2 Origin Host (Linux) Setup Requirements

7.2.1 Credentials

The RMM server requires SSH public key authentication (aka "passwordless ssh") for either the "root" user, or, via sudo, a non-root user, between itself and each Linux system to be migrated or replicated in order to enable various operational functions. If a user other than root is being used then that user must be set up as a user with full sudo privileges. In either case, the RMM's public ssh key (located at /root/.ssh/id_rsa.pub) needs to be appended to the appropriate authorized_keys file (/root/.ssh/authorized_keys or ~/.ssh/authorized_keys on each Linux source server being migrated.

If the root user is to be used by the RMM to access the source server, then passwordless ssh can be set up by running the command "ssh-copy-id <source server ip>". When the command is executed, it will prompt for the root password. Once this password is entered, the rmm's public ssh key will be appended automatically to the root authorized_keys file, /root/.ssh/authorized_keys. Passwordless ssh access as root will thus be completely set up automatically.

Verify passwordless ssh has been set up by running "ssh root@<source server ip address>" on the RMM.

If a user other than the root user is to be used by the RMM to access the source server, additional steps are needed.

7.2.1.1 Setup for user other than root

In the steps below it is assumed that the non-root user being used by the RMM to access the source server is "rackware". If you wish to set up a user other than "rackware", replace "rackware" with the desired username in all the steps below and in the sudo-config.txt file.

1. Login to your source host through SSH as root. If you can not log in as root, then log in as a user with full sudo privileges, and prepend 'sudo' to the commands on the source host shown below

2. Create the user "rackware" on the source host and make its default shell the bash shell.

```
# useradd -m -s /bin/bash rackware
```

3. Login to the RMM server and copy the sudoers information contained in /opt/rackware/docs/sudo-config.txt to the clipboard.

   [root@rmm] # cat /opt/rackware/docs/sudo-config.txt

4. Edit the sudoers file on the source host

   a) # visudo

   b) copy the contents of sudo-config.txt from RMM (via the clipboard) to the bottom of the sudoers file on the source host. Save the file.

5. Create .ssh directory and authorized_keys file for rackware user on source host

   a) # sudo su rackware

   b) $ mkdir -p /home/rackware/.ssh

   c) $ touch /home/rackware/.ssh/authorized_keys

   d) $ chmod 700 ~/.ssh/

   e) $ chmod 600 ~/.ssh/authorized_keys

6. Copy RMM SSH key to source host

   a) On the RMM server, run "cat /root/.ssh/id_rsa.pub"

   b) Copy the contents of that file to the clipboard. Copy only the contents of that file - do not include any extra spaces.

   c) Paste the contents of that file to the end of the source host's /home/rackware/.ssh/authorized_keys file

Once these steps are complete, test that passwordless ssh is working by running "ssh rackware@<source server ip address>" on the RMM. Verify that the ssh command is successful and that nothing that looks like an error message is shown when the ssh command is successful. For example, if running

```
[root@rmm-rackware ]# ssh rackware@<source ip address>
```

returns something like

```
tput: No value for $TERM and no -T specified
tput: No value for $TERM and no -T specified
tput: No value for $TERM and no -T specified
[Wed May 20 20:27:46 GMT 2023]
```

then a script may be running for non-interactive logins that is causing conflicts for the rackware user. Please check with the Linux System Administrator to determine how to prevent this type of message from being generated.

The contents of the sudo-config.txt file referenced above is:

```
# ---- BEGIN RACKWARE SUDOERS CONFIGURATION ----
# Append this file to /etc/sudoers
# Example:
# cat sudo-config.txt >> /etc/sudoers

User_Alias RW_MGMT_USERS = rackware
```

```
Runas_Alias RW_MGMT_RUNAS_USER = root

RW_MGMT_USERS ALL=(RW_MGMT_RUNAS_USER) NOPASSWD: ALL

Defaults:RW_MGMT_USERS !requiretty
# ---- END RACKWARE SUDOERS CONFIGURATION ----
```

7.2.2 Free Extents

The RMM uses LVM and Device Mapper to take snapshots of each partition. The RMM then copies the data from the snapshot rather than from the live file system.

By default, each Linux volume group must have a minimum 15% of the Used Space of the file systems on the volume group available as free extents for LVM snapshots. Issue a 'vgs' command (use 'sudo vgs' if logged in as a non-root user); for each volume group shown, the VFree value shows the amount of free extents in that volume group.

By default, if amount of free extents is less than 15% of Used Space of the file systems on a VG, the migration will fail before data transfer begins with error indicating how much additional space is needed. For example:

```
RSP_SYNC_ERROR  SYNC_ERROR_INSUFFICIENT_ORIGIN_FREE_EXTENTS  STATUS_ERROR
Details:
Some LVM VGs on the origin have insufficient free extents for snapshots.
VG Name            Add'l. Reqd.  For LV Snapshots
--------------------  ------------  ------------------------
centos             300 MB        var usr
data               68 MB         d01
```

Here's an example of volume groups with insufficient free extents:

```
[rackware@vr-centos79x64-03 ~]$ sudo vgs
  VG     #PV #LV #SN Attr   VSize   VFree
  centos   1   2   0 wz--n- <39.00g 4.00m
```

The free space in the volume group 'centos' is only 4 MB, which is less than 15% of the total size of that volume group.

To resolve this, an additional disk can be added to the source and the volume group can be extended. In our example, an additional disk was added, and the volume group 'centos' was extended to about 79 GB, resulting in approximately 40 GB of free space in the volume group:

```
[rackware@vr-centos79x64-03 ~]$ sudo vgs
  VG     #PV #LV #SN Attr   VSize  VFree
  centos   2   2   0 wz--n- 78.99g 40.00g
```

It is possible that a server will have an extremely high update rate, and thus will need more free extents. In that case additional free extents will need to be added to that server.

If there are environments where it is not possible to add the needed amount of free extents, and the use case is purely a migration use case, it may be possible to work around this requirement, as described in the explanation of the allow-direct-fscopy parameter in the 7.4 RMM GUI Operations Guide. Having free extents and syncing from a snapshot, however, is the preferred method for pure migration use cases. For Disaster Recovery use cases, no work around is possible, so adequate free extents must be in place on the source server.

7.2.3 Utilities

If the target machine will be PXE booted, and then the origin machine must have iscsi-initiator-utils installed.
If the target machine will be PXE booted, and the origin machine is running RHEL/OEL/Centos 7, SLES12, or RHEL/OEL/Centos 8 then the origin machine must also have dracut-network installed.
If the target machine will not be PXE booted, then there are no required utilities on the origin machine

The 'tar' utility must be installed on the Linux source servers. Some Centos 8 installations do not have it installed by default. Doing 'yum install tar' on those Centos 8 servers will install it.

7.2.4 Linux Antivirus Settings

If an antivirus product is deployed on the Linux origin/production servers, whitelist the directory /mnt/rackware/.

7.2.5 No noexec on /tmp and /var/tmp

The RMM needs to be able to execute files that are in /var/tmp/rackware by default The origin server must not have "noexec" on the lines in the /etc/fstab file that have any mounpoints for the /tmp/and /var/tmp directories. Additionally, be sure that any lines in /etc/fstab that reference subdirectories of /tmp and /var/tmp do not contain the "noexec" option. If the /etc/fstab file does have "noexec" on any of those lines, then remove the "noexec" from those lines and then remount the partitions using the commands:

```
sudo mount -o remount,exec /tmp
sudo mount -o remount,exec /var/tmp
```

If a security policy prevents removal of "noexec" from /var/tmp, the directory the RMM uses can be changed from /var/tmp to some other directory, which does not have a noexec option in the /etc/fstab file. To do this, open in an editor the file /opt/rackware/data/options on the RMM. Find the line that says

```
#policy_remoteTempDir=/var/tmp/rackware
```

Remove the "#" and change the line to specify a directory that does not have a noexec option, such as

```
policy_remoteTempDir=/mnt
```

or

```
policy_remoteTempDir=/media
```

Then save the file.

The RMM must be restarted for the changes to take effect. At a time when no syncs are in progress and no DR Policies are active, issue the command “rwadm restart”.

To see the value of an environment variable that is being used by the RMM at any given time, you can use “rw rmm show -v”, such as

```
rw rmm show -v | grep policy_remoteTempDir
```

7.2.6 Miscellaneous

The origin server must have an /etc/default/grub file on it. The origin server must use grub or grub2 as its bootloader, and have a working grub configuration, including having a valid /etc/default/grub file on it.

The origin server must have at least 20 MB of free space in the /var/tmp folder.

7.3 Origin Host (Windows) Setup Requirements

In order to perform various operations on Windows hosts, the RMM server requires a method to access the server. There are multiple connectivity options for Windows:

- SSH-only (best practice)
- Local User
- Domain User

It is considered best practice to use SSH-only for Windows access as it's the easiest, quickest, and has the least requirements/impact on the Target side. This document assumes that SSH-only is being used. Please contact RackWare if you wish to use a Local User or a Domain User rather than SSH-only as a method for the RMM to communicate with the Windows servers in the Origin environment.

It should also be understood if there are any GPO settings that will restrict RDP logins to the servers in the Target environment.

7.3.1 Volume Shadow Copy Service and Partition Free Space

The RMM utilizes the standard Windows VSS. Each VSS Writer on a Windows server must be in the Stable state whenever a sync is started.

Additionally, to support VSS, each Windows volume must have some free space to enable proper and complete snapshot execution using VSS. The amount of free space required is a function of update rate of a given volume. Technically if you want a rule that guarantees a successful sync in all cases, you'd need free space to be 100% of the used space. But typically, DBs require 15% to 20% and App and Web server take less. Many servers can be captured with much less. And high update rate servers, while rare, can require greater than 20%.

The best practice is to have snapshot space of 15% of the Used Space for each Windows volume. If syncs fail some of the time due to a snapshot getting deleted due to a lack of snapshot space, then increase the amount of snapshot space.

The free space that is required for snapshots is calculated on a per-volume basis, but the snapshot/shadow storage space for a given volume can reside on any volume. For example, the shadow copy storage space for the C drive could reside on the E drive.

Regardless of the snapshot settings, the C drive must always have at least 1 GB of free space.

7.3.2 Configuring Windows for SSH-Only

The RMM replicates and syncs Windows servers without requiring a windows password. This is referred to as SSH-Only. To properly configure SSH on Windows the RMM provides a small MSI that needs to be installed on the Origin server.

The RMM may ssh to the Windows server with 1 of 3 user types:

1. The “Administrator” user
2. The SYSTEM user (this is the user most commonly used by RackWare customers, and should be considered the default choice)
3. A user other than “Administrator” or the SYSTEM user; this user must be a Local Account of type Administrator

Before proceeding, determine which of those 3 you would like the RMM to use when it ssh’es to the Windows server.

Next, determine which user you will use to install the MSI package. This must be either “Administrator” or a user other than Administrator (#3 in the list above).

- If the RMM will be accessing the Windows server as "Administrator", then install the MSI while logged in to the Windows server as "Administrator" or as a user other than “Administrator” that has a Local Account of type Administrator.
- If the RMM will be accessing the Windows server as a user (with a Local Account type of Administrator) other than "Administrator" or the SYSTEM user, then install the MSI while logged in to the Windows server as that user; or install the MSI while logged in to the Windows server as Administrator; or install the MSI while logged in to the Windows server as a different user which has a Local Account of type Administrator.
- If the RMM will be accessing the Windows server as the SYSTEM user, then the MSI can be installed while logged in as either "Administrator" or while logged in as a user (with a Local Account type of Administrator) other than "Administrator".

*We recommend using the SYSTEM user because other accounts can present potential issues. For example, passwords may expire, accounts can get locked out, or be denied login. While it is possible to

use other accounts and we will continue to support this option, it is not the preferred method, especially for Disaster Recovery.

7.3.2.1 Adding the User in "Log on as a Service" Security Policy

The RackWare SSH service must be automatically started when the Windows server boots up. In order to have that happen, the user being used to install the MSI must be added to the "Log on as a Service" security policy using the following steps:

1. Go to Local Security Policy -> Local Policies -> User Rights Assignments
2. Search for 'Log on as a Service'
3. Click on Add User/Group, to add the desired user, either "Administrator" or the user other than Administrator which has Administrator privileges

7.3.2.2 Running the Rackware SSHD installer

The msi installer package can be run either from a web browser or directly on the Windows host.

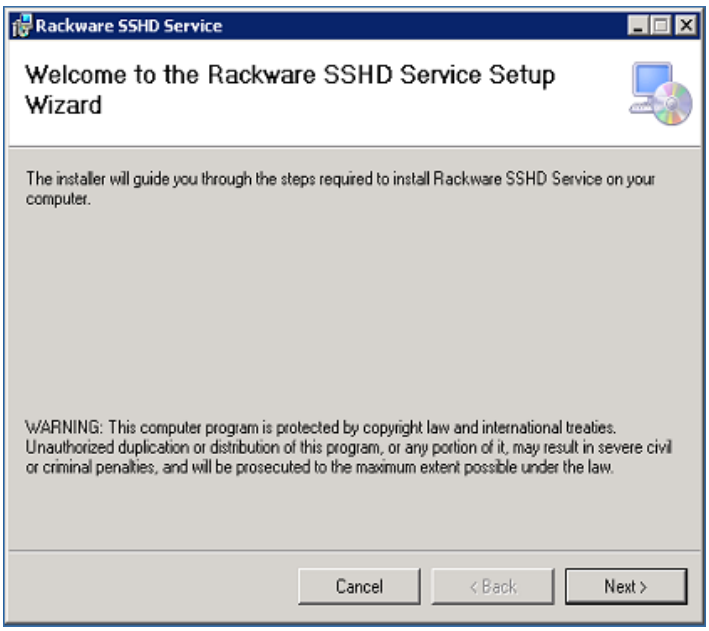
7.3.2.2.1 Running Installer from a Web Browser

First, log in to your Windows server as the user you have decided to use to install the MSI.

The MSI installer package can be run from a web browser (assuming your browser settings allow this) by entering one of the following URLs on your browser:

Windows 64-bit:
https://<your-RMM-IP-or-FQDN-Address>/windows/RWSSHDService_x64.msi
Windows 32-bit:
https://<your-RMM-IP-or-FQDN-Address>/windows/RWSSHDService.msi

Depending on the browser settings, this will either cause the setup wizard to be shown or will show the msi at the bottom of the browser window (in which case, right-click and choose 'Open' to bring up the Rackware SSHD Service Setup Wizard).

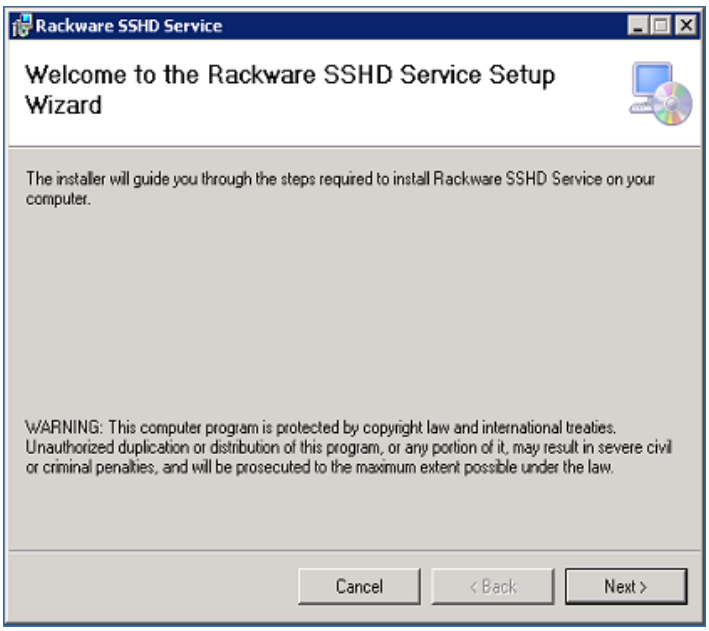


7.3.2.2.2 Running Installer from the Windows Host

The MSI's can be copied from the RMM server to the Windows host. The MSI's are at the following locations on the RMM:

1. ***For 64 bit**: /opt/rackware/utils/winbash/rwsshdservice-msi/amd64/RWSSHDService_x64.msi*
2. ***For 32 bit**: /opt/rackware/utils/winbash/rwsshdservice-msi/x86/RWSSHDService.msi*

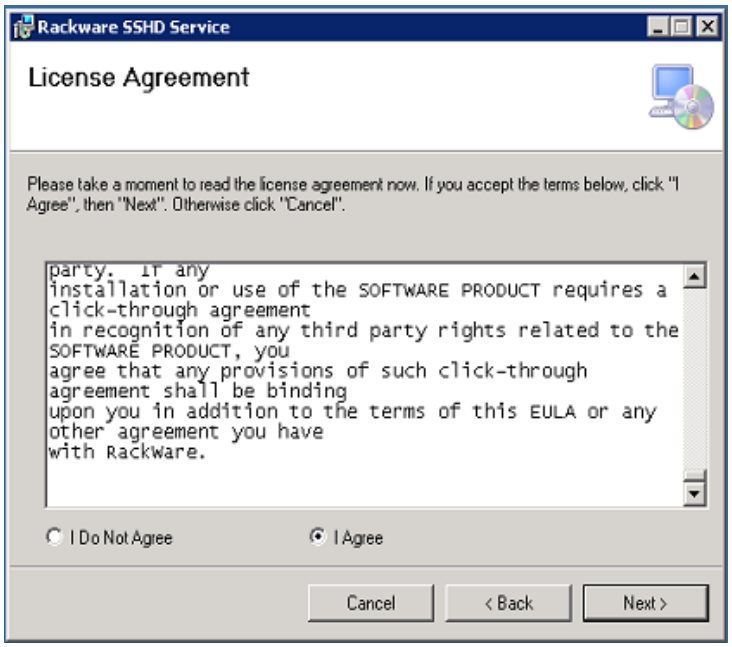
Once the appropriate file has been copied to the Windows host, log in to the Windows host as the user you have decided to use to install the MSI. Then double-click on the appropriate .msi file. This will cause the Rackware SSHD Service Setup Wizard to be shown.



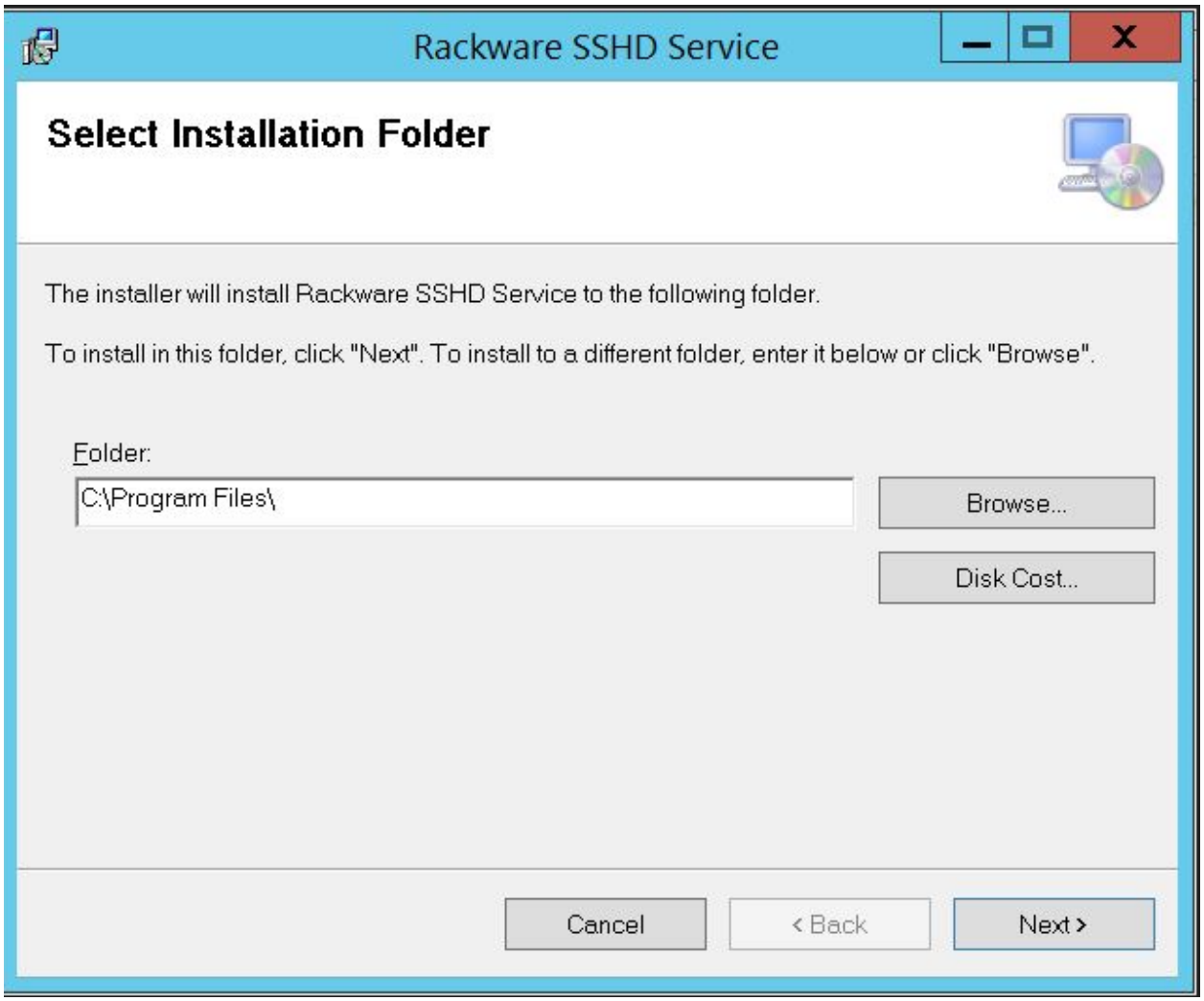
7.3.2.3 Steps to Install New MSI using the Installer

This section outlines the steps to install MSI for the first time.

1. Once the "Welcome to the Rackware SSHD Service Setup Wizard" page is displayed, press Next to begin the installation.
2. Read and accept the "License Agreement"

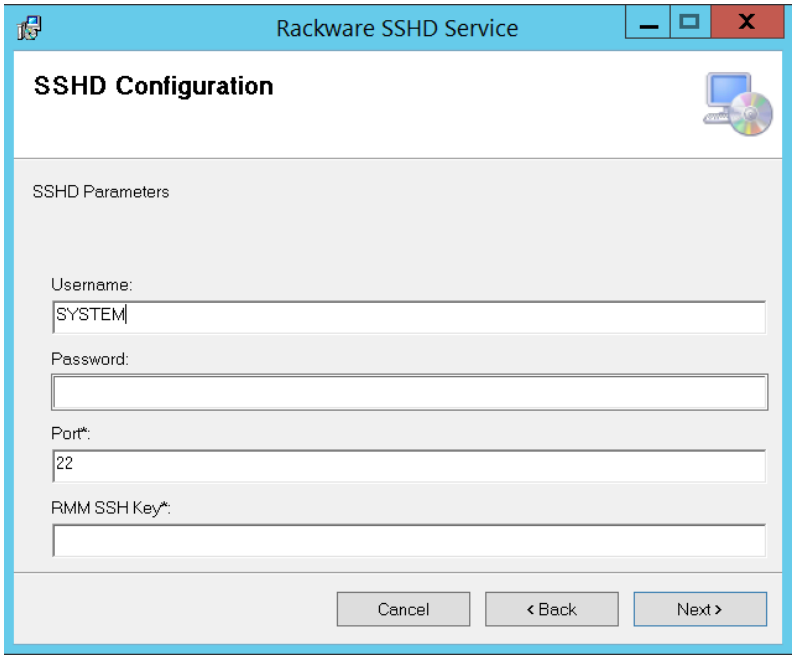


Press Next, and the Installation Folder window will be shown:



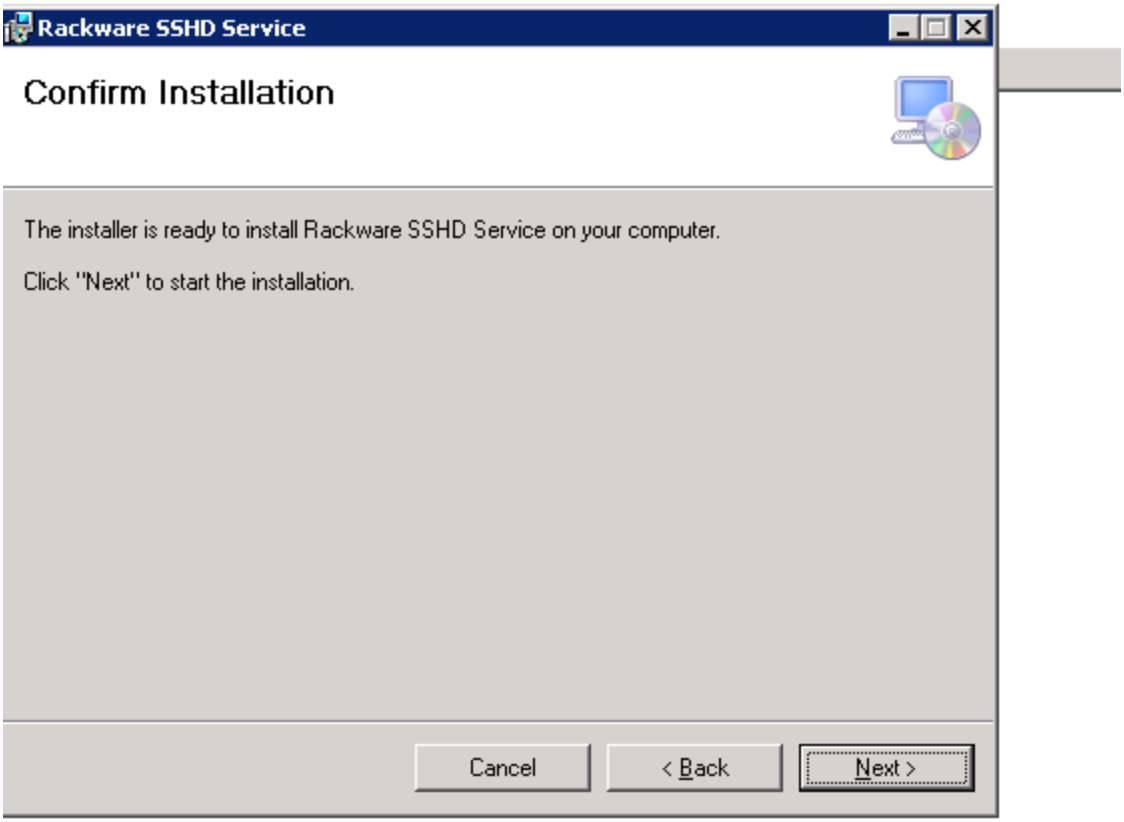
3. Click on 'Disk Cost' if you wish to see how many bytes the installation of the MSI will take, or the 'Browse' button if you wish to change the folder where the MSI will be installed. The default location for a 64-bit machine is C:\Program Files\, and the default location for a 32-bit machine is C:\Program Files(x86).

Then press the 'Next' button. The SSHD Configuration window will be shown.

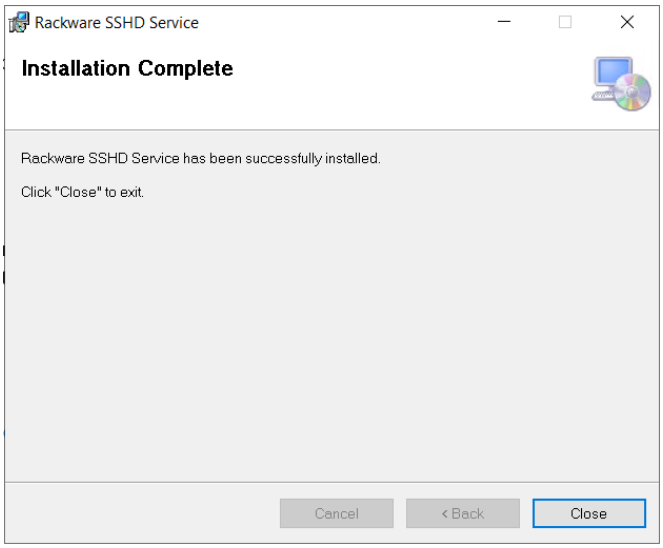


4. On the SSHD Configuration window:

   a. If the RMM will be accessing the Windows host as a user other than the SYSTEM user, then enter the appropriate username, either 'Administrator', or the non-Administrator, non-SYSTEM user. If the RMM will be accessing the Windows host as the SYSTEM user, then the username field should show SYSTEM or may be left blank.
   b. If the RMM will be accessing the Windows host as a user other than the SYSTEM user, then enter the appropriate password for the username that was entered. If the RMM will be accessing the Windows host as the SYSTEM user, then leave the Password field blank.
   c. Enter the port number you will be using for ssh. By default, this is 22. If you wish to use a custom ssh port (i.e. a port other than 22) as the ssh port, enter that number here. If a custom port is used, be sure to open that port on the appropriate firewalls. Note that the target server will use the same custom ssh port as the origin server.
   d. Enter the RMM's public key. This is the contents of the file /root/.ssh/id_rsa.pub on the RMM server. Be sure you can paste the entire key in one copy/paste operation, and that there are no /blanks/spaces pasted to the end of the key.
   e. Then press Next and the Confirm Installation screen will be shown



5. Press Next to begin the installation
6. After the installation completes, you will see the Installation Complete window.



Press the Close button.

After configuring all of the above steps for SSH-only, verify that the SSH public key authentication (aka “passwordless SSH”) is working by running the following command from the RMM:

"ssh <UserName>@< Windows Host IP Address>"

If using a custom SSH-port, then run the following command from the RMM:

“ssh <UserName>@< Windows Host IP Address>” -p <custom-ssh-port>

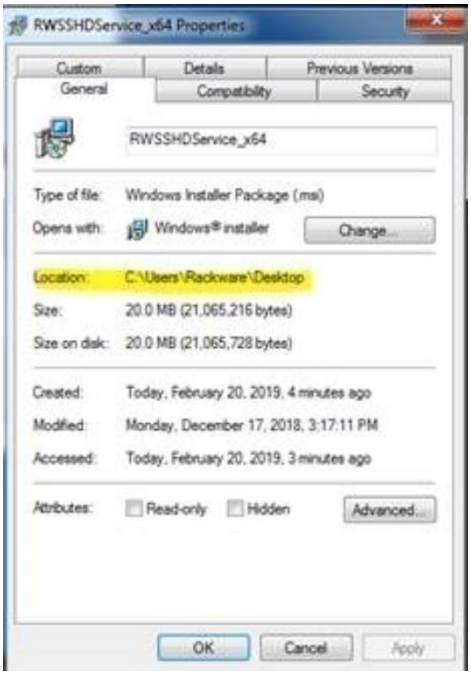
If it is working properly, you will be at a prompt on the Windows host.

Note that at no point in the installation process is a reboot of the Windows host required.

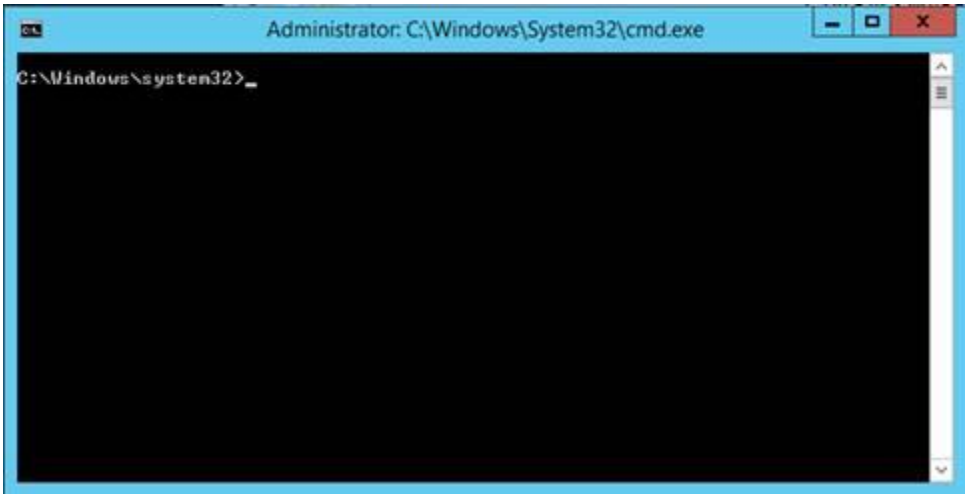
7.3.2.4 Steps to Install New MSI from the Windows Command Prompt

There is an option to install the new MSI from the Windows command prompt. Follow the preparations steps from section 7.3.2.2 to download the RackWare SSHD installer. Note the location where the installer was downloaded. To see the exact location right click on the MSI icon and select properties.

From there the actual path where the file is located can be noted:



Now access the command prompt from Windows.



For non-interactive installation, one of the following two modes should be specified as part of the command:

/passive (Unattended Mode - this option will show progress of the installation)
/quite (Quiet Mode – this option will not show progress of the installation)

Now type in the command using the following command format:

msiexec /passive /i [Path and Filename of the downloaded MSI] TARGETDIR=[Install Path] SVCUSERNAME=[User] PASSWORD=[Password] PORT=[Port] RMMSSHKEY=[ssh public key]

TARGETDIR=[Install Path] = full path of where the msi will be installed
SVCUSERNAME=[User] = Windows user with administrative privileges
PASSWORD=[Password] = Password for the user above
PORT=[Port] = The port to use for ssh. Enter 22 here for the default ssh port or another port number for a custom ssh port
RMMSSHKEY=[ssh public key] – This is the ssh public key from the RMM server. It is located in /<home dir of user used to install the RMM>/.ssh/id_rsa.pub.

For example, if using the Administrator userid:

msiexec /passive /i C:\Users\Rackware\Desktop\RWSSHDService_x64.msi TARGETDIR="C:\Program Files\" SVCUSERNAME="Administrator" PASSWORD="Winpass1" PORT="22" RMMSSHKEY="ssh-rsa AAAAB3NzaC1yc2EBBBBBIwAAAQEA64zieWE23z9fCqoq+VZ4m/Rq6WPBECyXxe1MZVIjwtfefV2RP3LIVS9DsIc7GOn4w7FzMZWL4HXV29d5Ccf4PJ0pd+8i6CscUmXFCwmDJp8dTEl+OuZEyNq5HLjDJzqCdih0ijhB5vjjDZVbdxEdzXKhHeL8sp+2rbsaWqkgE9WCgblrekrpJDvHp6mHOj8LCJkXU4qF+IJubUE6oTrSV9+fJSh9CSEmV/qOGK4rx2EeyzYE8+iIBAD+UQK4ilsvlX2rWIlEbayKyXq0m6bxsKSw2jG1bj0DAGwak3Af9e+S6SW7zMnu2FjrOY7Ib+3wkF8WnqXsmLCKKanSiMqvGQ== rmm@test-rmm01"

If an SVCUSERNAME is not specified, the SYSTEM user is used by default. Thus, if using the SYSTEM user, the msiexec command would be:

msiexec /passive /i C:\Users\Rackware\Desktop\RWSSHDService_x64.msi TARGETDIR="C:\Program Files \" SVCUSERNAME="" PASSWORD="" PORT="22" RMMSSHKEY="ssh-rsa
AAAAB3NzaC1yc2EBBBBBIwAAAQEA64zieWE23z9fCqoq+VZ4m/Rq6WPBECyXxe1MZVIjwtfefV2RP3LIVS9
DsIc7GOn4w7FzMZWL4HXV29d5Ccf4PJ0pd+8i6CscUmXFCwmDJp8dTEl+OuZEyNq5HLjDJzqCdih0ijhB5vjj
DZVbdxEdzXKhHeL8sp+2rbsaWqkgE9WCgblrekrpJDvHp6mHOj8LCJkXU4qF+IJubUE6oTrSV9+fJSh9CSEmV
/qOGK4rx2EeyzYE8+iIBAD+UQK4ilsvlX2rWIlEbayKyXq0m6bxsKSw2jG1bj0DAGwak3Af9e+S6SW7zMnu2Fj
rOY7Ib+3wkF8WnqXsmLCKKanSiMqvGQ== rmm@test-rmm01"

7.3.2.5 Checking for MSI Version

Once MSI is installed, to check the version of the installer, check the registry

HKEY_LOCAL_MACHINE -> SOFTWARE -> Rackware -> Version (Type= Reg_sz , Data = 2.0.0)

7.3.2.6 Uninstallation/Upgrade

Note - Before doing an uninstall, verify that the user being used to install/uninstall the MSI was added to the "Log on as a Service" security policy (see section 7.3.2.1). If this has not been done, uninstalling the Rackware SSHD service may cause it to become disabled, necessitating a reboot before the MSI can be installed successfully.

For changing/upgrading the MSI on a host, uninstall the old one by doing the following:

1. Log in to the Windows server with the same user you used when installing the MSI package.
2. Stop the RackWare SSHD service
3. Uninstall the Rackware SSH Service program via the Control Panel (Uninstall Programs)
4. Delete the C:\Program Files\Rackware-winutil folder (delete C:\Program Files (x86)\Rackware-wintuil if the 32-bit msi was installed).

Once the MSI has been uninstalled, follow the installation procedure to install the new MSI.

7.3.3 Windows Antivirus Settings

Some RMM operations can be inhibited or significantly slowed down by antivirus protections.

To avoid this, whitelist, that is, configure the antivirus software to not intercept and scan data that the following processes read:

- rsync.exe
- rwattr.exe
- rwchangesvc.exe
- rw_tngsync_util.exe
- rwchangedrv.sys

To avoid an antivirus program from terminating or removing any of the Rackware utilities associated with syncing a server, be sure the folder exclusion list /whitelist includes the following folders (assuming the Windows server is a 64bit server):

- C:\Windows\Temp\Rackware-winutil\
- C:\Program Files\Rackware-winutil\ (*1)
- C:\Program Files\RackWare Inc\RWSyncModule\
- C:\Windows\System32\rwchangesvc.exe
- C:\Windows\System32\Drivers\RwChangeDrv.sys

Windows 2016, 2019, and 2022 have built-in antivirus software (Windows Defender) that is enabled by default if no other AV software is installed. If the built-in AV software is being used, then the following processes should be listed in the Exclusions section of processes to be excluded:

- rsync.exe
- rwattr.exe
- rwchangesvc.exe
- rw_tngsync_util.exe
- rwchangedrv.sys

If the built-in AV software is being used, then the following folders should be whitelisted (assuming the Windows server is a 64bit server):

- C:\Windows\Temp\Rackware-winutil\
- C:\Program Files\Rackware-winutil\ (*1)
- C:\Program Files\RackWare Inc\RWSyncModule\
- C:\Windows\System32\rwchangesvc.exe
- C:\Windows\System32\Drivers\RwChangeDrv.sys

Note - Some AntiVirus programs may require the full path name of the process to be whitelisted. The full paths are as follows:

rsync.exe - %windir%\Temp\Rackware-winutil\bin\rsync.exe (*2)
rwattr.exe - %windir%\Temp\Rackware-winutil\bin\rwattr.exe
rwchangesvc.exe is %windir%\system32\ rwchangesvc.exe
rw_tngsync_util.exe - if a full path is needed then whitelist everything in the %windir%\Temp\RackWare-winutil\var\tmp\rackware\bin directory.

*1 – C:\Program Files is the default directory used when installing the RackWare MSI package on a 64-bit source server. If a different directory is used, then the Rackware-winutil subdirectory will be in that directory and should be on the whitelist.
*2 - %windir% is the value of the Windows windir environment variable; typically, this is C:\Windows, but it could be set to something else.

7.3.3.1 PowerShell Commands for Windows Defender

If using PowerShell commands to configure the exclusions for Windows Defender on a 64bit server, the commands are:

```
Add-MpPreference -ExclusionProcess "rsync.exe"
Add-MpPreference -ExclusionProcess "rwattr.exe"
Add-MpPreference -ExclusionProcess "rwchangesvc.exe"
Add-MpPreference -ExclusionProcess "rw_tngsync_util.exe"
Add-MpPreference -ExclusionProcess "rwchangedrv.sys"

Add-MpPreference -ExclusionPath “C:\Windows\Temp\Rackware-winutil\”
Add-MpPreference -ExclusionPath “C:\'Program Files'\Rackware-winutil\”
Add-MpPreference -ExclusionPath “C:\'Program Files\RackWare Inc\RWSyncModule'\”
Add-MpPreference -ExclusionPath “C:\Windows\System32\rwchangesvc.exe”
Add-MpPreference -ExclusionPath “C:\Windows\System32\Drivers\RwChangeDrv.sys”
```

If using PowerShell commands to verify the exclusions, the commands are:

```
Get-MpPreference | Select-Object -Property ExclusionProcess -ExpandProperty ExclusionProcess
Get-MpPreference | Select-Object -Property ExclusionPath -ExpandProperty ExclusionPath
```

7.3.4 Windows Language Support

The RMM currently supports Windows servers which have English as their default language. If your Windows servers do not have English as their default language, please see https://rackware.freshdesk.com/a/solutions/articles/5000890219 or contact RackWare Support for additional information.

8 Miscellaneous Notes

8.1 Minimum Memory

- When the RMM autoprovisions a server, by default the RMM will create the new server with the same amount of memory as the origin server. An exception to this is if the origin Windows server has less than 2 GB RAM. In that case the target server will have 2 GB RAM, rather than the amount the origin server had.